

Lecture 9 – Rao–Blackwellization

Last Modified: September 23, 2026

Outline:

9.1	Review: conditional expectation	1
9.2	Expectation conditional on a random variable	2
9.3	Rao–Blackwellization	3

9.1 Review: conditional expectation

Let (X, Y) be a continuous random vector. Recall that its joint density is $f_{X,Y}(x, y)$, its marginal density is

$$f_X(x) = \int f_{X,Y}(x, y) \, dy,$$

and the conditional density of X given $Y = y$ is

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

The conditional expectation of X and $g(X)$ given $Y = y$ is

$$\mathbb{E}[X | Y = y] = \int x f_{X|Y}(x | y) \, dx, \quad \mathbb{E}[g(X) | Y = y] = \int g(x) f_{X|Y}(x | y) \, dx.$$

The conditional variance of X given $Y = y$ is

$$\text{Var}(X | Y = y) = \mathbb{E}[(X - \mathbb{E}[X | Y = y])^2 | Y = y] = \mathbb{E}[X^2 | Y = y] - \mathbb{E}[X | Y = y]^2. \quad (9.1)$$

This is just the variance of the random variable with pdf $f_{X|Y}(x | y)$.

Example 1. Suppose $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Exp}(1)$ and let $Y = X_1 + X_2$. Our goal is to derive the conditional density of X_1 given $Y = y$. We know that $Y \sim \Gamma(2, 1)$, so

$$f_Y(y) = ye^{-y} \mathbb{1}_{\{y \geq 0\}}.$$

We first find the conditional density of Y given $X_1 = x$. Once $X_1 = x$ is fixed, $Y = x + X_2 = x + \text{Exp}(1)$. Thus Y is in a location family, and we worked out before how to get its density:

$$f_{Y|X_1}(y | x) = e^{-(y-x)} \mathbb{1}_{\{y-x \geq 0\}} = e^{-(y-x)} \mathbb{1}_{\{y \geq x\}}.$$

We can now obtain the joint density by multiplying the conditional density by the marginal density of X_1 :

$$\begin{aligned} f_{X_1,Y}(x, y) &= f_{Y|X_1}(y | x) f_{X_1}(x) \\ &= e^{-(y-x)} e^{-x} \mathbb{1}_{\{x \geq 0, y \geq x\}} \\ &= e^{-y} \mathbb{1}_{\{0 \leq x \leq y\}}. \end{aligned}$$

It follows that

$$f_{X_1|Y}(x | y) = \frac{f_{X_1,Y}(x, y)}{f_Y(y)} = \frac{e^{-y}}{ye^{-y}} \mathbb{1}_{\{0 \leq x \leq y\}} = \frac{1}{y} \mathbb{1}_{\{0 \leq x \leq y\}}.$$

Thus,

$$X_1 \mid Y = y \sim \text{Unif}(0, y),$$

which gives

$$\mathbb{E}[X_1 \mid Y = y] = \frac{1}{2}y.$$

To compute e.g. $\mathbb{E}[X_1^2 \mid Y = y]$, we take an integral:

$$\mathbb{E}[X_1^2 \mid Y = y] = \int_0^y x^2 \frac{1}{y} dx = \frac{1}{3}y^2.$$

The conditional variance is then

$$\text{Var}(X_1 \mid Y = y) = \mathbb{E}[X_1^2 \mid Y = y] - \mathbb{E}[X_1 \mid Y = y]^2 = \frac{1}{12}y^2.$$

9.2 Expectation conditional on a random variable

Definition 1 (Expectation and variance conditional on a random variable). Let Z be any random variable, for example $Z = X$ or $Z = g(X)$. For each possible y , let $m(y) = \mathbb{E}[Z \mid Y = y]$ and $v(y) = \text{Var}(Z \mid Y = y)$, defined as in (9.1). Then we define $\mathbb{E}[Z \mid Y]$ to be the random variable $m(Y)$, and $\text{Var}(Z \mid Y)$ to be the random variable $v(Y)$. Equivalently, the variance conditional on Y is

$$\begin{aligned} \text{Var}(X \mid Y) &= \mathbb{E}[X^2 \mid Y] - \mathbb{E}[X \mid Y]^2 \\ &= \mathbb{E}[(X - \mathbb{E}[X \mid Y])^2 \mid Y] \end{aligned} \tag{9.2}$$

Usually, expectations and variances are deterministic numbers. But when you take the expectation or variance conditional on a random variable, the result is itself a random variable. Let's see how this works in the context of Example 1. We computed that $\mathbb{E}[X_1 \mid Y = y] = \frac{1}{2}y$, $\mathbb{E}[X_1^2 \mid Y = y] = y^2/3$, and $\text{Var}(X_1 \mid Y = y) = \frac{1}{12}y^2$. Therefore,

$$\mathbb{E}[X_1 \mid Y] = \frac{1}{2}Y, \quad \mathbb{E}[X_1^2 \mid Y] = \frac{1}{3}Y^2, \quad \text{Var}(X_1 \mid Y) = \frac{1}{12}Y^2, \quad Y \sim \text{Gamma}(2, 1).$$

Theorem 1 (Tower property and total variance). Let (X, Y) be a random vector, with X having finite second moment. Then:

- (i) The **tower property** (or law of total expectation) is

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].$$

- (ii) The **law of total variance** is

$$\text{Var}(X) = \text{Var}(\mathbb{E}[X \mid Y]) + \mathbb{E}[\text{Var}(X \mid Y)].$$

The tower property says that averaging in two stages gives the same overall mean. The law of total variance splits the variation in X into

- variation of the conditional averages across different values of Y , and
- variation of X among outcomes with the same value of Y .

In particular,

$$\text{Var}(\mathbb{E}[X | Y]) = \text{Var}(X) - \underbrace{\mathbb{E}[\text{Var}(X | Y)]}_{\text{variation averaged out}} \leq \text{Var}(X).$$

Proof of Tower property. We do it for continuous random variables; for discrete random variables, change the integral to a sum. The key is to remember that $\mathbb{E}[X | Y] = m(Y)$ is just some function of Y (“ m ” for mean), so we can apply LOTUS:

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X | Y]] &= \mathbb{E}[m(Y)] = \int m(y) f_Y(y) dy \\ &= \int \mathbb{E}[X | Y = y] f_Y(y) dy \\ &= \int \left(\int x f_{X|Y}(x | y) dx \right) f_Y(y) dy \\ &= \iint x f_{X,Y}(x, y) dx dy \\ &= \int x f_X(x) dx = \mathbb{E}[X]. \end{aligned}$$

In the second-to-last line, we used that $f_{X|Y}(x | y) f_Y(y) = f_{X,Y}(x, y)$. □

Proof of the total-variance identity (optional). Write $m(Y) = \mathbb{E}[X | Y]$. Then

$$\begin{aligned} \text{Var}(\mathbb{E}[X | Y]) &= \text{Var}(m(Y)) = \mathbb{E}[m(Y)^2] - \mathbb{E}[m(Y)]^2 \\ &= \mathbb{E}[m(Y)^2] - \mathbb{E}[\mathbb{E}[X | Y]]^2 = \mathbb{E}[m(Y)^2] - \mathbb{E}[X]^2, \end{aligned} \tag{9.3}$$

using the tower property in the last step. Next, using (9.2), we have $\text{Var}(X | Y) = \mathbb{E}[X^2 | Y] - m(Y)^2$, so that

$$\mathbb{E}[\text{Var}(X | Y)] = \mathbb{E}[\mathbb{E}[X^2 | Y]] - \mathbb{E}[m(Y)^2] = \mathbb{E}[X^2] - \mathbb{E}[m(Y)^2]. \tag{9.4}$$

Adding (9.3) and (9.4) gives

$$\begin{aligned} \text{Var}(\mathbb{E}[X | Y]) + \mathbb{E}[\text{Var}(X | Y)] &= (\mathbb{E}[m(Y)^2] - \mathbb{E}[X]^2) + (\mathbb{E}[X^2] - \mathbb{E}[m(Y)^2]) \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \text{Var}(X). \end{aligned}$$

Conditional on Y , add and subtract $m(Y)$:

$$\begin{aligned} \mathbb{E}[X^2 | Y] &= \mathbb{E}[(X - m(Y) + m(Y))^2 | Y] \\ &= \mathbb{E}[(X - m(Y))^2 | Y] + m(Y)^2 \\ &= \text{Var}(X | Y) + \mathbb{E}[X | Y]^2. \end{aligned}$$

□

9.3 Rao–Blackwellization

Recall that we are looking for an unbiased estimator whose variance is smallest out of all the unbiased estimators, for all possible values of θ (the UMVUE). Towards that goal, we start with a simpler one: If U is an unbiased estimator of $g(\theta)$, how can we systematically construct another unbiased estimator with smaller variance?

The answer is *conditioning*: start with an unbiased U , and condition on another statistic S . The two key observations are:

1. **Averaging reduces variance and preserves unbiasedness.** For any statistic S , the law of total variance gives

$$\text{Var}_\theta(\mathbb{E}_\theta[U | S]) \leq \text{Var}_\theta(U).$$

The law of total expectation shows that this averaging preserves the mean, so if $\mathbb{E}_\theta[U] = g(\theta)$, then $\mathbb{E}[\mathbb{E}_\theta[U | S]] = g(\theta)$ as well.

2. **The averaged estimator must not depend on the unknown parameter.** For an arbitrary statistic S , the conditional expectation $\mathbb{E}_\theta[U | S]$ may have the form $h_\theta(S)$ and therefore may not be computable from the data alone. If instead we condition on a *sufficient* statistic T , then the conditional distribution of $\underline{X}$ given T does not depend on θ . Consequently, for every statistic $U = U(\underline{X})$,

$$\mathbb{E}_\theta[U | T] = h(T) = h(T(\underline{X}))$$

for a function h that does not depend on θ . Thus this new estimator can be computed solely from the given sample.

This is the crux of Rao–Blackwellization: conditional averaging gives variance reduction, and sufficiency ensures the result is actually a statistic (a function of the data alone).

Theorem 2 (Rao–Blackwell). Let $U = U(\underline{X})$ have finite second moment, be unbiased for $g(\theta)$, and let $T = T(\underline{X})$ be sufficient for θ . Define

$$U^* = \mathbb{E}_\theta[U | T].$$

Then:

- (i) U^* is a statistic: it is a function of T that does not depend on the unknown θ ;
- (ii) $\mathbb{E}_\theta[U^*] = \mathbb{E}_\theta[U] = g(\theta)$ for every θ ;
- (iii) $\text{Var}_\theta(U^*) \leq \text{Var}_\theta(U)$ for every θ , and $\text{Var}_\theta(U^*) = \text{Var}_\theta(U)$ holds if and only if U is a function of T .
- (iv) $\text{MSE}_{U^*}(g(\theta)) \leq \text{MSE}_U(g(\theta))$ for every θ , with strict inequality whenever U is not a function of T .

The theorem says we are guaranteed MSE reduction for any U that is not already a function of T .

Proof. Because T is sufficient, the conditional distribution of $\underline{X}$ given T does not depend on θ . Since U is a function of $\underline{X}$, the conditional distribution of U given T also does not depend on θ . Therefore $U^* = \mathbb{E}_\theta[U | T]$ can be calculated as a function of T without knowing θ .

The law of total expectation gives

$$\mathbb{E}_\theta[U^*] = \mathbb{E}_\theta[\mathbb{E}_\theta[U | T]] = \mathbb{E}_\theta[U].$$

Thus U and U^* have the same bias. The law of total variance gives

$$\text{Var}_\theta(U) = \text{Var}_\theta(U^*) + \mathbb{E}_\theta[\text{Var}_\theta(U | T)] \geq \text{Var}_\theta(U^*). \quad (9.5)$$

Combining the same bias with the smaller variance proves the MSE statement. $\square$

Variance equality proof (optional). Equality of variances in (9.5) holds exactly when $\text{Var}_\theta(U | T) = 0$ with probability one. But recalling the definition of conditional variance in (9.2), we see that $\text{Var}_\theta(U | T) = 0$ if and only if $U = \mathbb{E}_\theta[U | T] = U^*$ with probability one. Finally, let us show $U = \mathbb{E}_\theta[U | T]$ if and only if U is a function of T . If $U = \mathbb{E}_\theta[U | T]$, then U is automatically a function of T . Conversely, if $U = h(T)$, then for each t , we have $\mathbb{E}[h(T) | T = t] = h(t)$, so $U^* = \mathbb{E}[h(T) | T] = h(T) = U$. $\square$

Example 2 (Rao-Blackwellization for Poissons). Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta), \quad n \geq 2,$$

and we want to estimate $g(\theta) = \mathbb{P}(\text{Poisson}(\theta) = 0) = e^{-\theta}$. The natural estimator is the proportion of the samples X_i which are equal to 0:

$$U = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i=0\}}$$

Exercise: check U is unbiased! Now, we know that $T = \sum_{i=1}^n X_i$ is sufficient for θ . Thus the Rao-Blackwellized estimator for U is $U^* = \mathbb{E}[U | T]$. Let's write $U^* = h(T)$, where $h(t) = \mathbb{E}[U | T = t]$, and compute this function. Notice that we didn't write $\mathbb{E}_\theta[U | T = t]$, because we know this expectation won't depend on θ by sufficiency of T . We have

$$\begin{aligned} h(t) &= \mathbb{E}[U | T = t] \\ &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i=0\}} \mid T = t\right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\mathbb{1}_{\{X_i=0\}} \mid T = t] \\ &= \mathbb{E}[\mathbb{1}_{\{X_1=0\}} \mid T = t] \\ &= \mathbb{P}(X_1 = 0 \mid T = t). \end{aligned} \tag{9.6}$$

To derive the conditional probability of $X_1 = 0$ given $T = t$, write

$$T = X_1 + \sum_{i=2}^n X_i,$$

where X_1 is independent of $\sum_{i=2}^n X_i$. We have

$$\begin{aligned} \mathbb{P}(X_1 = 0 \mid T = t) &= \frac{\mathbb{P}_\theta(X_1 = 0, X_1 + \sum_{i=2}^n X_i = t)}{\mathbb{P}_\theta(T = t)} \\ &= \frac{\mathbb{P}_\theta(X_1 = 0, \sum_{i=2}^n X_i = t)}{\mathbb{P}_\theta(T = t)} \\ &= \frac{\mathbb{P}_\theta(X_1 = 0) \mathbb{P}_\theta(\sum_{i=2}^n X_i = t)}{\mathbb{P}_\theta(T = t)} \end{aligned}$$

Next, note that $X_1 \sim \text{Poisson}(\theta)$, $\sum_{i=2}^n X_i \sim \text{Poisson}((n-1)\theta)$, and $T = \sum_{i=1}^n X_i \sim \text{Poisson}(n\theta)$. Therefore, continuing from above,

$$h(t) = \mathbb{P}(X_1 = 0 \mid T = t) = \frac{\mathbb{P}(\text{Poisson}(\theta) = 0) \mathbb{P}(\text{Poisson}((n-1)\theta) = t)}{\mathbb{P}(\text{Poisson}(n\theta) = t)}$$

$$= \frac{e^{-\theta} e^{-(n-1)\theta} (n-1)\theta^t / t!}{e^{-n\theta} (n\theta)^t / t!} = \left(1 - \frac{1}{n}\right)^t.$$

Therefore,

$$U^* = h(T) = \mathbb{P}(X_1 = 0 \mid T) = \left(1 - \frac{1}{n}\right)^T.$$

We know U^* is unbiased for $g(\theta) = e^{-\theta}$, and since the original U was not a function of T , the variance of U^* is strictly smaller than that of U . Thus the MSE of U^* is strictly smaller than the MSE of U . Note that U^* has a nice interpretation. We were trying to estimate $\mathbb{P}_\theta(X_1 = 0)$. We don't have access to this, but instead, we can compute $\mathbb{P}(X_1 = 0 \mid T)$, i.e. the conditional probability of one of the observations being zero given the total count $T = \sum_{i=1}^n X_i$.

Let us compare U^* with the plug-in estimator. The MLE of θ is

$$\hat{\theta}_{\text{MLE}} = \bar{X}_n = \frac{T}{n},$$

so the plug-in estimator (MLE for $g(\theta) = e^{-\theta}$) is

$$g(\hat{\theta}_{\text{MLE}}) = e^{-\bar{X}_n} = e^{-T/n} = (e^{-1/n})^T.$$

When $n \gg 1$, the two are close:

$$(e^{-1/n})^T \approx \left(1 - \frac{1}{n}\right)^T = U^*$$

by a Taylor expansion of $e^{-1/n}$. But for small n , they could be significantly different. In particular, one can show that $e^{-T/n}$ is biased, whereas we know by design that $U^* = (1 - 1/n)^T$ is unbiased.